

LAB TASK- 1 [C# Type Conversion]

Create a struct named **Product** along with the attributes & methods mentioned below:

Attributes:

- **Id** [Only Positive Number]
Data Type: Integral Type
- **Name**
Data Type: String
- **Category**
Data Type: String
- **Price**
Data Type: Floating Type
- **StockQuantity**
Data Type: Integral Type
- **SupplierEmail**
Data Type: String
- **Rating**
Data Type: Floating Type (scale 1-5)
- **DiscountPercentage**
Data Type: Floating Type (Percentage)

Methods:

1. **Constructor**
Functionality: A parameterized constructor that accepts all the attributes as input and initializes them in the struct.
2. **FullProductInfo**
Return Type: String
Functionality: Return a concatenation of product name, category, and price.
3. **PriceAfterDiscount**
Return Type: Floating Type

Functionality:

- Calculate the discount amount in price: [Discount Percentage * Price]
- Deduct the discount from the original price and return the new price after applying the discount.

4. **RatingStatus**

Return Type: String

Functionality: Return a status based on the rating:

- "Excellent" for rating 4.5 and above
- "Good" for rating between 3.0 and 4.4
- "Average" for rating between 1.0 and 2.9
- "Poor" for rating below 1.0

Step - 1

Use Console.ReadLine() to ask for the following input from the user in a Console Application:

Id, Name, Category, Price, StockQuantity, SupplierEmail, Rating, DiscountPercentage

- Perform the necessary type conversions and pass all type-casted input parameters into the struct constructor to create a struct object.
- While collecting information from the user, if a user provides invalid information (i.e., fails the type conversion), assign default values to the attributes.
- Make the first letter of **Name** and **Category** capitalized (uppercase) after input from the user.

Step - 2

- Call the available functions using the struct object and display the full product information, the price after applying the discount, and the rating status.